

Principles for Managed Entry Agreements in the WHO European Region

policy framework



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Abstract

Since 2023, the WHO Regional Office for Europe's Access to Novel Medicines Platform (NMP) has brought together public and private stakeholders to improve equitable access to high-cost, novel medicines. Working Group 3 focuses on developing principles for payment, pricing, health technology assessment and reimbursement that support the sustainability of both health systems and industry. A key output, the set of managed entry agreement (MEA) principles, was developed collaboratively by WHO/Europe, National Institute for Health and Care Excellence, Utrecht University and experts from NMP Work Package 3 between March 2024 and June 2025. The process included literature reviews, focus groups, interviews and expert feedback. The MEA principles aim to guide countries and health technology developers in managing uncertainty about the financial impact and performance of new medicines. They are voluntary and were designed to be updated biannually. The framework includes (i) eight core MEA principles (two are specific to outcome-based agreements), (ii) an uncertainty scale, (iii) flow chart and (iv) checklist. These tools help to visualize the decision-making process, clarify the phases of MEAs and promote transparency. Ultimately, the MEA principles support countries to refine or establish processes for successful MEA implementation through multistakeholder collaboration and sharing best practices.

Keywords

PHARMACOECONOMICS, COSTS AND COST ANALYSIS – pricing, DRUG COSTS, TECHNOLOGY ASSESSMENT, BIOMEDICAL, ECONOMICS, PHARMACEUTICAL

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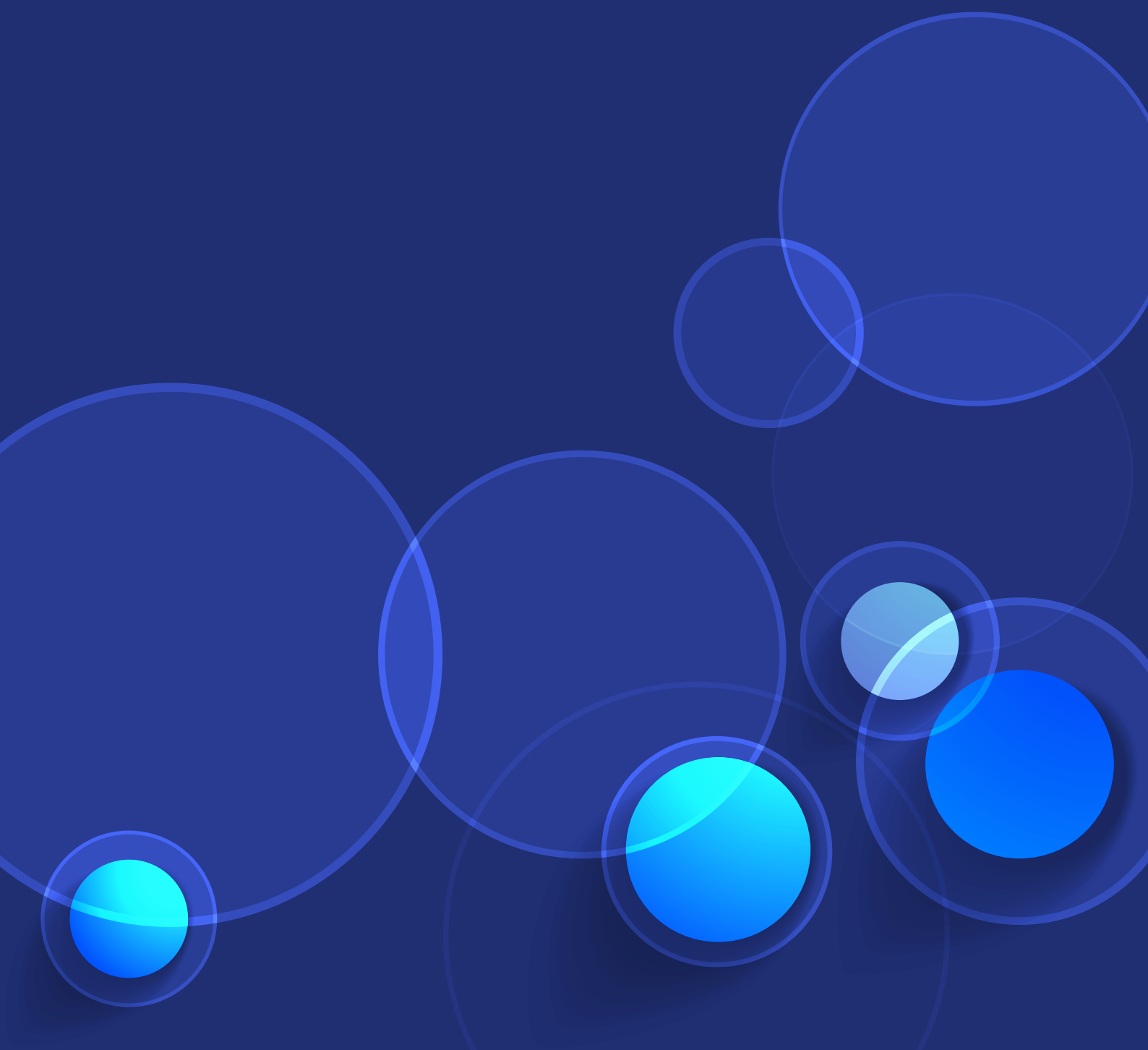
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Abbreviations

AI	Artificial Intelligence
CDF	Cancer Drug Fund (United Kingdom)
EPAR	European Public Assessment Reports
e-prescribing	electronic prescribing
HTA	health technology assessment
IMF	Innovative Medicines Fund
MAA	managed access agreement
MEA	managed entry agreement
NHS	National Health Service (United Kingdom)
NICE	National Institute for Health and Care Excellence (United Kingdom)
NMP	WHO Regional Office for Europe's Access to Novel Medicines Platform
PAS	patient access scheme
VT	<i>Voorwaardelijk toegelaten zorg</i> [conditionally approved care] (Netherlands (Kingdom of the))
ZIN	<i>Zorginstituut Nederland</i> [National Health Care Institute] (Netherlands (Kingdom of the))

Introduction



Background

The WHO Regional Office for Europe's Access to Novel Medicines Platform

The WHO European Programme of Work 2020–2025: United Action for Better Health (1) highlights the importance of equitable and sustainable access to high-quality medicines to achieve universal health coverage and meet the United Nations Sustainable Development Goals (2). On 14 March 2023 in Almaty, Kazakhstan, the Standing Committee of the Regional Committee's Working Group on Governance approved the establishment of WHO Regional Office for Europe's Access to Novel Medicines Platform (NMP). This unique multistakeholder platform was created following the conclusion of the Oslo Medicines Initiative of 2020–2022 (3).

Through the NMP, the Regional Office will act as a neutral convenor to enable State and non-State actors (including private sector and nongovernmental organizations, academic institutions, and philanthropic foundations) to agree on and jointly deliver solutions to improve affordable and equitable patient access to effective, novel, high-cost medicines.

Overview of NMP working groups

The NMP has established four working groups:

- Working Group 1 on transparency focuses on improving the transparency of and access to information throughout the life-cycle of medicines;
- Working Group 2 on solidarity focuses on fostering structured dialogue between stakeholders and collaboration for the accrual, evaluation and use of evidence of clinical and economic value across the life-cycle of these medicines;
- Working Group 3 on sustainability focuses on developing access principles for novel treatments, including guidance on horizon scanning and managed entry agreements (MEAs); and
- Working Group 4 on novel antimicrobials focuses on a needs assessment for improving access and aligning eligibility criteria for incentives.

Working Group 3 on sustainability aims to develop principles (including for payment, pricing, health technology assessment and reimbursement) that recognize the need for sustainability of health systems and private industry and to enable risk sharing and good governance of markets, including

ways to address market failures and unmet needs. In this context, an MEA is defined as an arrangement between a health technology developer (or any other legal parties acting on behalf of the health technology developer such as registered commercial wholesalers) and payer/provider that enables access to or coverage/reimbursement of a health technology subject to specified conditions for the given therapy (4,5). These arrangements aim to mitigate the evidentiary and financial uncertainty about treatments or identify situations where access (coverage/reimbursement) is not possible at the proposed price. A variety of mechanisms are used to manage the adoption of technologies to maximize their effective use and impact on a payer's budget while guaranteeing sustainable commercial terms for health technology developers.

The classification of the different types of MEAs was initially based on terms developed by the Organization for Economic Co-operation and Development (4) and other relevant literature (6–14). It is important to mention that within the NMP, Working Group 1 on transparency¹, the DEFINE subgroup is collating all relevant definitions, including those for MEAs. The following list of key terms will be updated to be consistent with those included in the DEFINE project, when finalized.

Financial-based MEAs include:

- **discounts/rebates** – price reductions granted to specified purchasers under specific conditions prior to purchase;
- **price-volume agreements** – pricing agreements in which drug prices are progressively lowered or capped as more patients receive the treatment; and
- **subscription models** – reimbursement strategies that decouple payments from sales volumes by offering pharmaceutical manufacturers a fixed monetary amount regardless of how many units of the drug are sold.

Outcome-based MEAs include the following.

- **Coverage with evidence development** provides provisional reimbursement of promising technologies with limited clinical evidence, that is, temporary reimbursement is granted with an obligation for the manufacturer to obtain and provide additional data in either a clinical or a research setting. This can be organized such that patients either only have access when included in the study or have unrestricted access, with an obligation to generate data for that group of patients.
- **Conditional treatment continuation**, in which continuation of coverage for individual

¹ Working Group 1 on transparency comprised the following three subgroups: DEFINE, REFLECT, and EXPECT.

patients is conditional on meeting short-term treatment goals such that coverage ends when the agreed conditions are not met.

- **Pay for outcome**, through which the level of reimbursement is linked to the future performance of the product in a real-world environment such that therapy costs are fully or partly covered by the manufacturer if outcomes are not achieved. There is also an option to review ongoing reimbursement for the entire patient cohort if clinical outcomes are not being delivered.

In general, all outcome-based agreements include a financial agreement, for example, a discount. Note that there may be a variety of practical options for transacting reimbursement for each outcome-based MEA model, for example, by spreading payments over a period of time and subscription models. The model used is determined by the financial rules and legal requirements applicable to each country.

MEAs along the drug life-cycle

MEAs are tools to support health-care decisions and can be used to facilitate the uptake of novel medicines by health systems. Their design can be implicitly or explicitly based on a number of policy goals, such as uptake of innovation, sustainable financing of the health system, improvement of patient access and investment into value for money (15). The two most common types are financial-based MEAs and outcome-based MEAs. Broadly, financial-based MEAs focus on controlling the total expenditure of payers (i.e. budget impact of novel medicines) and outcome-based MEAs provide performance-related conditions for reimbursement or support real-world data collection to reduce uncertainties related to effectiveness (7–9,16). Successful implementation of MEAs requires a national legal basis, with details on timelines and responsibilities. The *WHO guideline on country pharmaceutical pricing policies* should be consulted for specific pricing considerations (17).

Financial-based MEAs are widely used in both higher- and lower-income countries across the WHO European Region (7,10).

Outcome-based MEAs are needed because regulatory approval is often expedited for novel medicines for diseases with a high unmet need. This leads to uncertainties related to long-term clinical benefits and real-world effectiveness when such medicines are launched. Such uncertainty makes it difficult for payers to commit to payment models based on future outcomes because they cannot reliably predict the long-term value of the treatments. Furthermore, novel treatments –

particularly those that are highly specialized (such as advanced therapy medicinal products), curative or used across multiple indications – pose funding challenges owing to their high upfront costs and the possibility that their long-term benefits may extend beyond immediate health-care savings. Therefore, outcome-based MEAs are a useful tool for mitigating uncertainty about the benefits of novel medicines. Such outcomes can be collected on an individual basis, with payment depending on the patient's outcome, or on a collective basis, whereby evidence gathered for a patient cohort is used to inform reassessment and new pricing and reimbursement decisions. Compared with financial-based MEAs, outcome-based MEAs have a more limited use because the measurement of outcomes is more complex than simply applying price discounts. Frequently mentioned challenges of outcome-based MEAs are as follows (15).

- The implementation and execution of MEAs (and, especially, outcome-based MEAs) come with a high administrative burden for all stakeholders.
- Many health systems lack the necessary data infrastructure – such as patient registries, electronic health records and payer databases – to support the implementation of outcome-based MEAs, which require data of sufficient quality/quantity to inform reassessment decisions.
- Legal and regulatory frameworks vary across countries, and are not always well suited to setting up outcome-based MEAs.
- Misaligned interests, disagreement and lack of coordinated efforts between stakeholders can hinder the development and execution of outcome-based MEAs.
- Lack of transparency can lead to mistrust among stakeholders and limit the potential for shared learning and improvement. Balancing the need for transparency with the requirement to maintain confidentiality of commercial terms is a significant challenge.
- There is often a lack of willingness to change and reluctance to adopt new models owing to risk aversion and limited experience with these approaches.

In addition, evidence on the successful implementation of outcome-based MEAs is mixed and shows that they can be costly to set up and difficult to implement (4,8,18,19). Evidence from the implementation of an outcome-based MEA in the United Kingdom that was used to mitigate uncertainty surrounding the clinical benefits of a novel multiple sclerosis medicine shows that MEAs can fail to link prices to outcomes, as intended (11).

A recent study in Italy found limited evidence that MEAs are effective in lowering total pharmaceutical expenditure (12). In contrast, other studies have highlighted that outcome-based MEAs may provide opportunities, including earlier access, reduced budget unpredictability and mitigation of opportunity costs (12,20,21).

Hence, some stakeholders consider that MEAs offer potential benefits for selected products by providing timely patient access to innovative treatments while supporting the sustainability of the health system and providing sufficient incentives for innovation. To reach the goal of achieving a win-win-win situation for all stakeholders, agreement is needed on a structured process for implementing MEAs, including guidance on how best to match a product with an MEA type (22,23). Hence, there is a need to develop widely accepted MEA principles.

MEA principles: methods and aim

Development of the MEA principles was a collaborative effort led by the WHO Regional Office for Europe, the United Kingdom’s National Institute for Health and Care Excellence (NICE) and Utrecht University and experts from NMP Work Package 3. All NMP members had the opportunity to provide feedback on drafts of the MEA principles. Fig. 1 gives a detailed overview of the multi-method approach taken to develop the MEA principles. In parallel, two country case studies (on Netherlands (Kingdom of the) and the United Kingdom) were drafted following the structure of the MEA principles, based on national expert opinions. These are included at the end of this report.

A need to develop MEA principles – from early access to benefits to society – was expressed by all members of Working Group 3 at a face-to-face meeting in Manchester, United Kingdom on 12–13 March 2024. Achieving this would

require evidence development and knowledge transfer between stakeholders both within and between countries so that all could benefit without duplication of efforts. Participants agreed that, while it may not be feasible to share the details of specific MEAs, a set of principles should be jointly developed to which all Member States in the WHO European Region can sign, as well as standardized templates on how to operate MEAs. After the consultation, stakeholders agreed to replace standardized templates with a checklist approach because MEA templates are individualized according to national legislative and regulatory frameworks. The checklist provides guidance to enable informed decision-making. Participants identified a series of questions to use as a framework for developing a set of principles for MEAs, as follows.

- How can the quality and effectiveness of MEAs be improved?
- When should MEAs be used by health systems?
- What types of MEA are most useful for health systems?
- What lessons can be learned from the perspectives of countries, industry and patients?
- Can these types of MEAs be made more transparent with regard to their set-up, design and implementation?
- Can some agreed principles/frameworks for these MEAS be developed, and how can capacity-building in the WHO European Region be supported?

At the Working Group 3 virtual meeting in May 2024 and NMP plenary meeting in July 2024, NMP members again expressed interest in taking forward the development of MEA principles, which led to this report being the commissioned.

Between September 2024 and December 2024, a core research team including members of the WHO NMP Secretariat, NICE and Utrecht University, developed the first version of the MEA principles based on a pragmatic literature search, document analysis and in-depth interviews with country

Fig. 1. Overview of the methodological steps



WG3: Working Group 3.

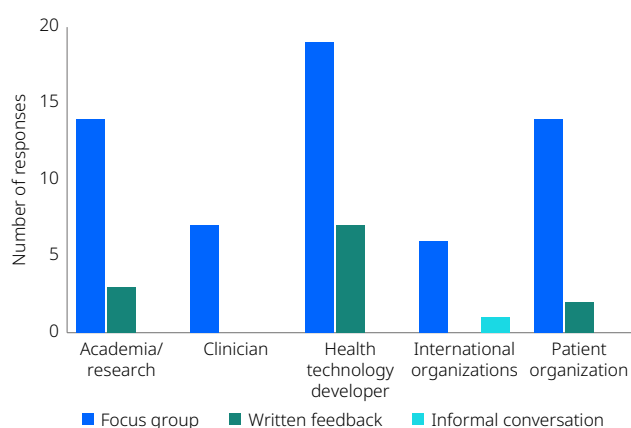
experts. Specifically, the literature search was based on a PhD thesis from Utrecht University (15) and other relevant literature (7,16,24–30).

From January to March 2025, a large-scale deliberation process on the first version of MEA principles took place: it included an online expert consultation facilitated by the RWE4Decisions network (31), as well as several interviews with country experts from the health technology assessment (HTA) and payer fields. This process replaced the initially planned Delphi process, which could not be executed owing to lack of funding and timing restrictions. Annex 1 lists the organizations that participated in the deliberation process.

From April to May 2025, the core research team incorporated the feedback and further developed the MEA principles. On 30 April 2025 all NMP members were briefed on the status of all NMP projects, including the MEA principles. Following this, NMP members and members of the RWE4Decisions network were invited to comment on the draft version of the document.

In total, representatives of 73 organizations participated in the focus group discussion or informal stakeholder conversation or provided written feedback on the second draft. This included representatives of 30 national public bodies (including regulators, HTA agencies and payer organizations/ministry of health), 13 international/national patient organizations, eight research organizations and four international organizations, as well as 12 health technology developers and six clinicians or registry holders. Representatives of some organizations provided feedback on multiple occasions, resulting in 81 individual responses (Fig. 2). Annex 2 presents a high-level summary of all received feedback from both rounds.

Fig. 2. Types of stakeholders who provided feedback



For an additional independent review, three academic experts were invited to provide feedback: Associate Professor Huseyin Naci from the London School of Economics, United Kingdom (first draft); Professor Steve Morgan, from the University of British Columbia, Canada (second draft); and Dr Beth Woods from the University of York, United Kingdom (second draft). All experts completed confidentiality agreements.

In summary, several key themes emerged during the deliberation phase, as follows.

- **Transparency:** industry representatives were initially hesitant about certain elements of the transparency proposal owing to concerns about maintaining confidentiality on pricing and proprietary information. Through conversations, the core team clarified that transparency only related to the design, process, implementation and data-sharing aspects of MEAs, with the aim to maintain stakeholder trust and facilitate coordination in collecting real-world evidence. HTA bodies and other stakeholder groups welcomed the transparency proposal.
- **Redesign of presenting results:** stakeholders of the round-table meetings recommended redesigning the MEA scale of uncertainty because the initial traffic light visual risked misinterpretation as a binary go/no-go system. This would promote financial-based MEAs over outcome-based MEAs, which differ for each drug or indication.
- **Managed exit:** HTA bodies highlighted the need to define transparent, structured and enforceable strategies during contracting to ensure that patient treatment continues regardless of MEA outcomes. Procurement agencies echoed this view and also underlined the role of clear exit frameworks as a tool to assist in managing clinical and economic expectation throughout the MEA. It was emphasized that this would require agreement on the achievable end-points during the contracting phase.

Aims of the MEA principles

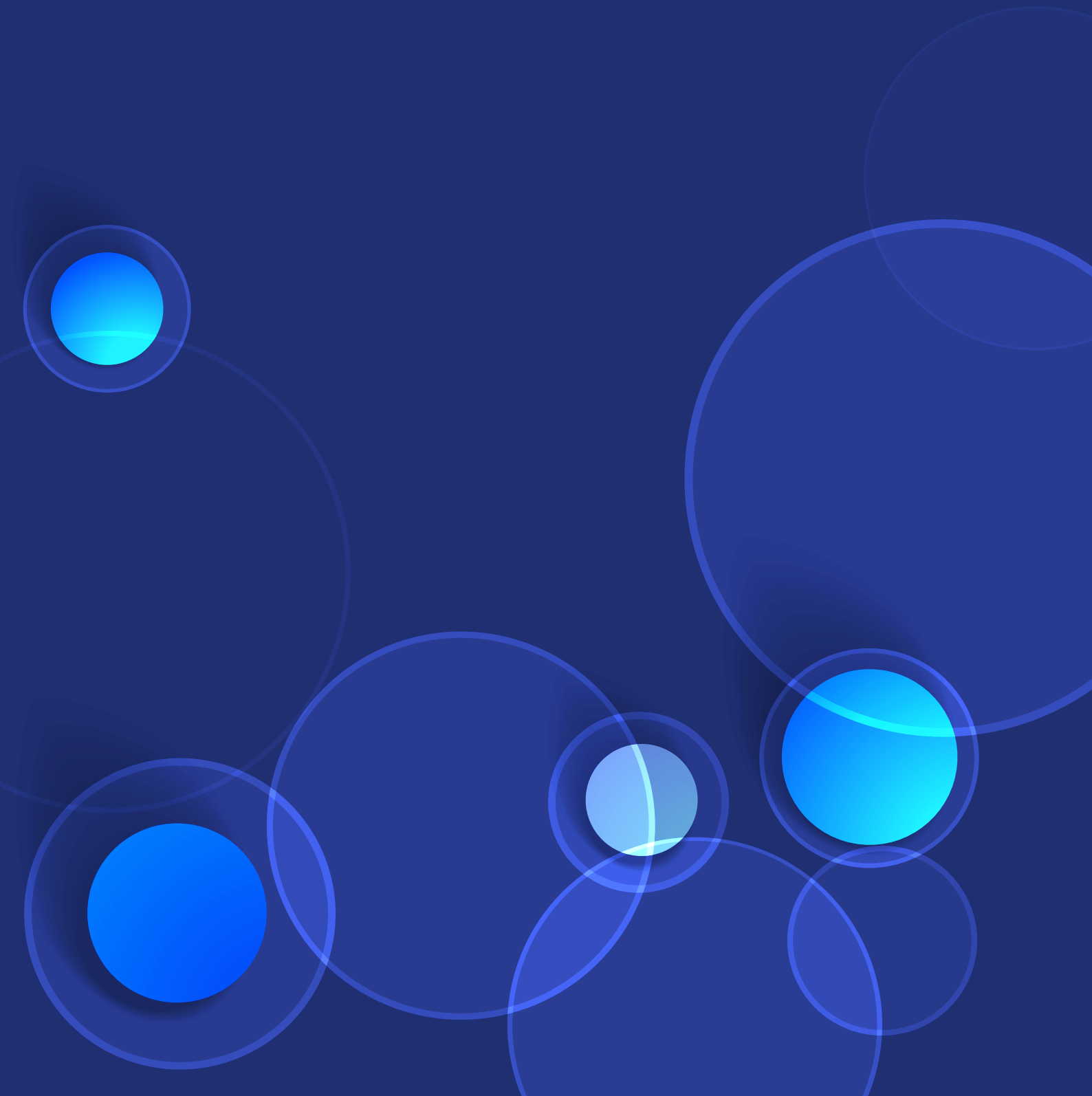
The overall aim of an MEA is to conclude an agreement between a health technology developer and health-care payers on the coverage of a new medicine while managing uncertainty around their financial impact or performance (4). With this in mind, the specific aim of the MEA principles is to present a set of core concepts and best practice examples of MEAs for countries to use to review their own processes. The principles may give guidance to all stakeholders to ensure that each step is considered systematically, an

element is not forgotten and that there is improved transparency where feasible. Specifically, for outcome-based MEAs these principles may help to navigate the difficult choice of selecting only those innovations that deliver overall societal benefits by considering the effects on access, uncertainties and expenditure (i.e. health opportunity costs) compared with standard treatment options. Use of the MEA principles is voluntary. They are intended to form a living document that reflects regular (biannual) updates in scientific knowledge on MEAs and rapid methodological developments for post-launch evidence generation that can be used in HTA/pricing policy-making. In addition the principles can help stakeholders to decide when MEAs are desirable (and when they are not).

Using the MEA principles

In recognition that European health systems differ depending on their funding (e.g. social insurance, national health system, mixed systems) and organizational structure (integrated, national systems versus regionally operated systems), application of the MEA principles will also vary. The MEA principles consist of four parts that should be conducted in the following order: MEA principles, MEA complexity scale, MEA flow chart and MEA checklist. Due to country-specific differences, not all parts or phases may be applicable to all contexts but they may be considered for critical decision-making and prioritization.

MEA principles



Before developing a MEA, several key principles should be considered: six general ones and two specifically for outcome-based MEAs.

General MEA principles are as follows.

- **Open and transparent HTA and reimbursement:** the starting point should always be the regular national policy route, which is based on open and transparent HTA and pricing policies. An MEA should only be considered if an HTA is not feasible.
- **Simplicity is often preferable:** MEAs should be as simple as possible to minimize the burden on the health system, including all stakeholders. Complex arrangements should only be considered once simple arrangements (which facilitate the fastest access) have been demonstrated to be unsuitable. Where there is a case for a more complex MEA, this need to be carefully scrutinized to ensure that complex monitoring or burdensome data collection and disproportionate additional costs are avoided.
- **Type of MEA depends on uncertainties:** although MEAs should be as simple as possible, their complexity should be proportionate to the specific uncertainties being addressed. For highly innovative therapies with complex value propositions, outcome-based agreements may be the most appropriate initial approach, rather than first requiring a demonstration that simpler arrangements are unsuitable.
- **Clear and agreed assessment and exit terms:** as part of the contracting phase, all stakeholders should agree on the terms of assessment, time frame and exit strategy. The latter should include consideration of different possible scenarios such as the data confirming the trial results (i.e. continued reimbursement), the data contradicting pivotal trial results (i.e. renegotiations) and inconclusive data (i.e. in the case of discontinuation, funding available for enrolled patients to continue treatment until it is clinically appropriate for them to stop).
- **Collaborative effort:** while respecting each stakeholder's remit, all relevant parties (including HTA bodies, payer organizations, industry, health-care providers and patient representatives) should collaborate to ensure that MEA development balances their diverse perspectives and does not delay access to innovative treatments.
- **Regular evaluation:** from the health system perspective, regular evaluation of the implemented MEAs is necessary, including information on whether conditions have been met within the agreed time frames and an estimation of required resources from all stakeholder perspectives.

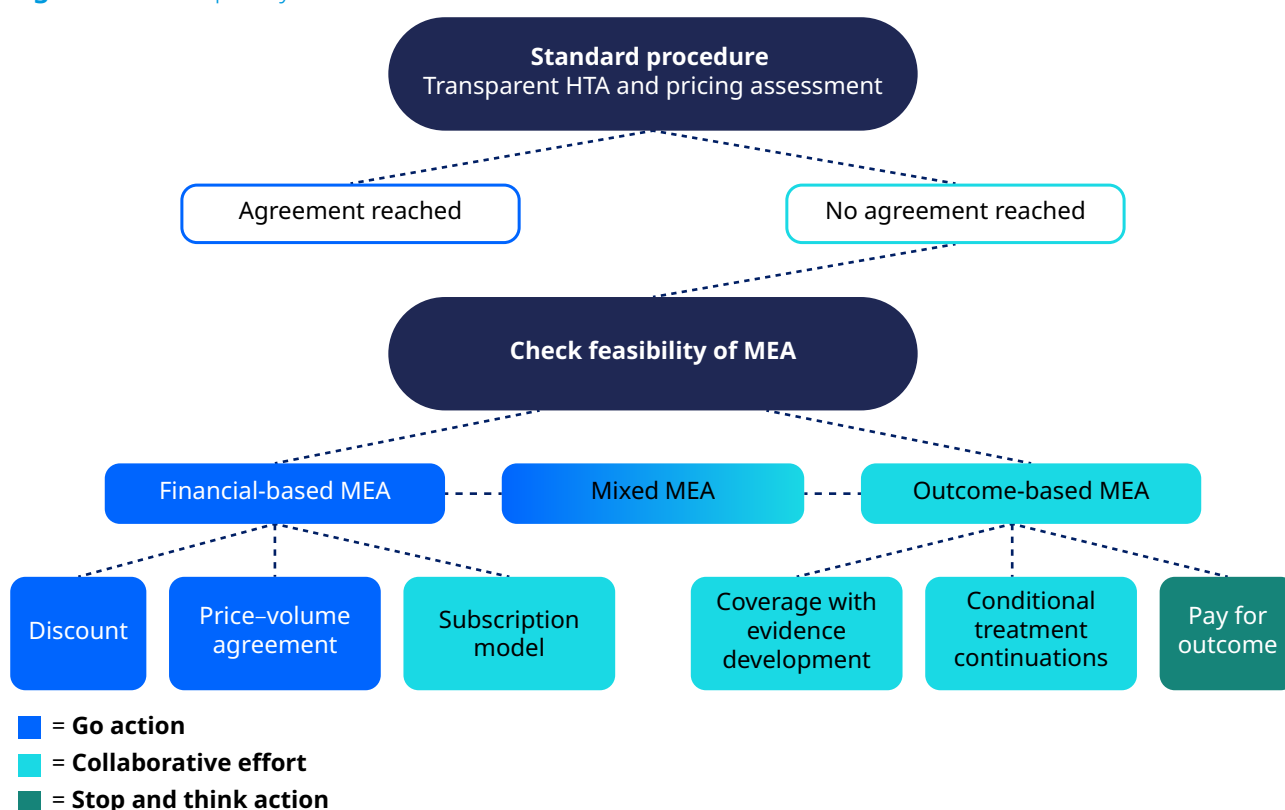
Specific principles for outcome-based MEAs are as follows.

- **Use of existing data structures is preferable:** to minimize the burden on health systems, existing data structures (including ongoing clinical trials, national registries and datasets) should be leveraged where appropriate. When existing structures are insufficient to capture relevant outcomes, stakeholders should collaborate to develop efficient new data collection methods, with consideration of possible industry contribution to data infrastructure development.
- **Aim for transparency:** transparency is recommended on the MEA type and agreed clinical outcome variables to mitigate uncertainties, in line with European Union joint clinical assessments and efforts to create a European Health Data Space (32). However, there should be a balance between appropriate confidentiality of commercial terms to enable sustainable market access strategies across countries with different economic conditions and the public interest, especially regarding the final set price.

MEA complexity scale

The MEA principles need to be considered from the perspective of national HTA and pricing/reimbursement decision-making processes. The MEA complexity scale shows the different branches of the decision-tree, including a colour scheme indicating the need for different types of MEAs depending on the uncertainty level around a drug's financial impact or performance (Fig. 3).

Notably, the MEA principles only focus on decisions for which no agreement could be reached during the standard, national HTA and pricing procedure. If a positive agreement was reached, then standard national pricing procedures apply; in some cases, these may include discounts and price-volume agreements. Please consult *WHO guideline on country pharmaceutical pricing policies* for more information (17).

Fig. 3. MEA complexity scale

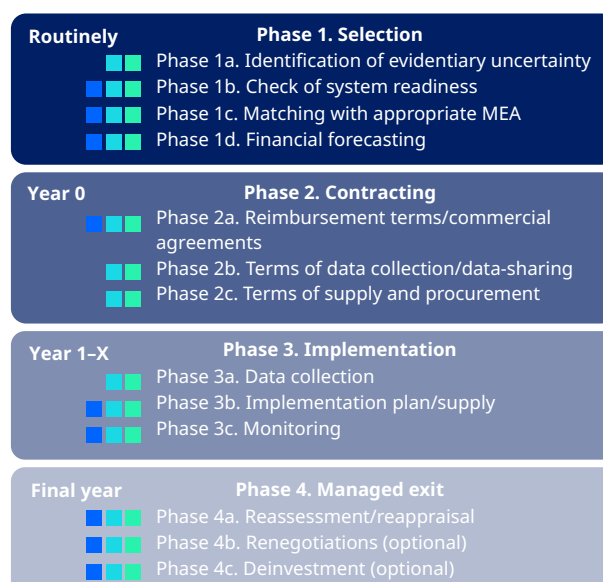
Notes: go action – low uncertainty (drug's effectiveness and financial impact) and low complexity of agreement; collaborative effort – increased uncertainty (drug's effectiveness and financial impact) and increased complexity of agreement; and stop and think action – high uncertainty (drug's effectiveness and financial impact) and high complexity of agreement.

MEA flow chart

The MEA flow chart summarizes all relevant phases that should be considered when checking the feasibility and relevant implementation phases of an MEA (Fig. 4). It also shows which phases should be done regularly and which are specific to the implementation phase of the MEA.

MEA checklist

The MEA checklist is based on the MEA flow chart and provides details on what is needed in each phase: points to agree on, stakeholders, sources of evidence, transparency and available tools/templates (Table 1).

Fig. 4. MEA flow chart

Note: the colour scheme is the same as for the MEA complexity scale and indicates which steps are needed for each MEA type.

Table 1. MEA checklist

Phase	Action
1. Selection (conduct regularly) Aim: to identify new products based on their decision-relevant uncertainties (either financial or evidentiary) and match them with appropriate MEAs	
1a. Identification of evidentiary uncertainty ■ ■	Criteria ☐ Checking regulatory status (e.g. conditional versus full marketing authorization) ☐ Outcomes of HTA ☐ Assessment of uncertainty level of evidence ☐ Previous line of treatment ☐ Number of competitors ☐ Clinical Pathway Factors a. Predicted patient numbers b. Market share c. Likelihood of competition d. Unmet need e. Current indications and revenues f. Budget impact Stakeholders Initiated by: ☐ health technology developer (as part of the reimbursement dossier) Involved: ☐ HTA agencies ☐ Ministry of Health ☐ clinical associations ☐ payer organizations Sources of evidence ☐ Reimbursement dossier and EU-JCA dossier (based on clinical trial data) ☐ 'Horizon scanning' documents ☐ EPAR (separately) ☐ Published literature (e.g. Taams et al. (33), Bloem et al. (34)) Transparency ☐ HTA agency/payer/Ministry of Health to publish list of selected products and whether an MEA has been used and what Templates/tools ☐ Annex 3 ☐ Horizon scanning, including perceived budget impact assessments a. At national level: examples from the Netherlands (Kingdom of the) (35) or the United Kingdom (36) b. At European level: HTACG subgroup for the identification of emerging health technologies (37) c. At international level: International Horizon scanning Initiative (38,39) ☐ Early and joint scientific consultations a. At European level: EUHTAR (40) b. At national level: example from the Netherlands (Kingdom of the) (41)

Phase	Action	
1b. Check of system readiness 	Criteria	☐ HTA agency/Ministry of Health to understand feasibility and time frame of possible data collection, including an overview of potential available sources of RWD and maturation of ongoing studies/new clinical studies ☐ Check availability of data collection (e.g. ongoing trials, access to registry data) - Which outcome variables/end-points are collected? - What is the timing of the data availability? - How accurate and complete are the available data? ☐ Check system readiness – is new technology disruptive to routine care? - Capacity (human resources) - Extra costs (e.g. prior genome sequencing required) - Extra handling (e.g. storage) ☐ Check whether compassionate use, early access or named patient programmes are in place ☐ Check appropriate use - Limitation to specific patient population - Controlled access programmes (e.g. vaccination check prior to administration of new treatment)
	Stakeholders	Initiated by: ☐ HTA agency ☐ Ministry of Health Involved: ☐ clinical associations ☐ HTD
	Sources of evidence	☐ Published literature (such as Brinkhuis et al. (42))
	Transparency	☐ Publication of criteria to assess
	Templates/tools	☐ Annex 3
1c. Matching products with appropriate MEA 	Criteria	☐ Guidance on the scoping of evidence (product-specific information on disease and treatment (PICO) (43)) ☐ Cost-effectiveness analysis/budget impact ☐ Value of information analysis (more relevant for HTA agencies than for payer organizations) (44)
	Stakeholders	Initiated by: ☐ payer organizations ☐ HTA agencies Involved: ☐ health technology developer
	Sources of evidence	☐ Reimbursement dossiers (HTA or HTD)
	Transparency	☐ HTA agency/payer/Ministry of Health to publish list of selected products and type of MEA ☐ Communication with clinical associations and patient organization

Phase	Action	
1d. Financial forecasting ■ ■ ■	Templates/tools	☐ Fair Pricing Calculator (45) ☐ MEA framework and calculation tool (46) ☐ Impact HTA checklist (47)
	Literature	☐ EU HI-PRIX project (48)
	Criteria	☐ Horizon scanning (if not nationally available look up sources from IHSI, NICE, ZIN) ☐ Existing data sources (e.g. registries) on national epidemiology of specific disease incl. information on potentially eligible patient population
	Stakeholders	☐ Payer organizations/Ministry of Health
	Sources of evidence	☐ Horizon scanning ☐ Early scientific advice (however confidential)
	Transparency	☐ Payer organization published annual budget
	Templates/tools	☐ National budget impact assessment as part of horizon scanning ☐ Possibly reports from the EU HTAR subgroup on horizon scanning

2. Contracting (year 0)

Aim: to develop a clear understanding among all stakeholders of the financial/commercial terms including managed exit and the terms with regard to data collection and supply

2a. Reimbursement terms/commercial agreements ■ ■ ■	Agreement on	☐ Terms of financial risk sharing based on: a. predicted patient numbers (required information) b. market share (required information) c. budget impact (required information) d. agreement on the consequences of alternative treatments entering the market e. unmet need (additional information) f. current and future indications and revenues (additional information) ☐ Payment terms (overview of different payment terms in HI-PRIX pay for innovation observatory) ☐ Timeline of the agreement ☐ Terms of exiting the agreement
	Stakeholders	Initiated by: ☐ health technology developer to Ministry of Health or payer organization Involved: ☐ HTA agencies ☐ health technology developer
	Sources of evidence	☐ Published literature ☐ Reimbursement dossier ☐ National available budget ☐ HTA cost-effectiveness models with different scenarios ☐ Economic modeling
	Transparency	☐ Sharing of data depends on the individual data-sharing agreements between the parties but should strive for transparency

Phase	Action	
	Templates/tools	☐ Results of national HTAs ☐ EU ASCERTAIN project (49) and its revision of the AIM fair price calculator (45) ☐ EU HI-PRIX project (48)
2b. Terms of data collection/data-sharing ■ ■	Agreement on	☐ End-points to be collected (e.g. response rate, duration of response, serious adverse events, overall survival, quality of life – preference given to validated, non-observer-dependent and patient-rel-evant end-points) ☐ Time frame of data collection (in case of longer contracts, mid-term reassessment suggested) ☐ Payment of data management ☐ Patient recruitment (prediction of patient number needed depends on the disease area) ☐ Data ownership ☐ For observational studies, ethical approval of protocol ☐ Patient consent ☐ Publication rights ☐ Statistical Analysis Plan ☐ Funding of data collection ☐ Rights to “access” and share data between involved stakeholders and external stakeholders (aggregated versus individual level data – especially sensitive for rare diseases and depersonalization of clinical data) ☐ Assessment of system readiness to collect data (burden on system, IT infrastructure, clinical staff) ☐ Data sources, collection processes, data quality metrics and methodologies for data analysis
	Stakeholders	Initiated by: ☐ hospitals/centres of excellence Involved: ☐ clinicians ☐ patients (reflection on outcomes and schedule to collect data) ☐ hospital pharmacist ☐ other health-care workers (e.g. physiotherapist, nurse, psychologist) ☐ data specialists
	Sources of evidence	Real-world evidence from: ☐ National clinical datasets ☐ Ongoing clinical trials ☐ Registry data ☐ Health technology developer
	Transparency	☐ Registration of clinical trial or link to registry ☐ Publication of agreed outcome variables to be collected and publication of data analysis ☐ Example from NICE: regular stakeholder meetings with involved stakeholders to check on data collection progress
	Templates/tools	☐ Templates for setting up different contracts on data collection (47) ☐ WHO policy and implementation guidance: sharing and reuse of health-related data for research purposes (50)

Phase	Action	
2c. Terms of supply and procurement ■ ■	Agreement on	Depending on structure of the health system: ☐ national procurement (if not integrated/national system, then preferably national procurement for rare diseases), including price-volume agreement or discount ☐ supply directly to hospital (negotiation of additional rebate) ☐ guaranteed supply for predicted patient numbers ☐ Payment flow
	Stakeholders	Initiated by: ☐ national procurement agency ☐ Ministry of Health/payer organization ☐ hospital pharmacy ☐ distributors/wholesalers Involved: ☐ health technology developers
	Sources of evidence	Not applicable
	Transparency	☐ Sharing of terms of supply and procurement depends on individual agreements between the parties but should strive for transparency
	Templates/tools	☐ No templates available
3. Implementation (year 1–X)		
Aim: to implement the MEA in a transparent way that all involved stakeholders are aware of their part of the agreement		
3a. Data collection ■ ■	Adherence to	☐ Agreed terms of data collection
	Stakeholders	Initiated by: ☐ clinicians ☐ other (independent) outcome assessor Involved: ☐ patients ☐ hospital pharmacists ☐ other health-care workers (e.g. physiotherapist, nurse, psychologist) ☐ IT department ☐ registry staff ☐ payer/ HTA oversight staff
	Sources of evidence	Not applicable
	Transparency	☐ Peer-review publication of clinical end-points (respecting confidentiality of patient information) ☐ Publication of HTA reports/HTA re-assessment reports (if applicable) ☐ Reasons why certain evidence was not collected
	Templates/tools	☐ Example from Spain: Valtermed (51)

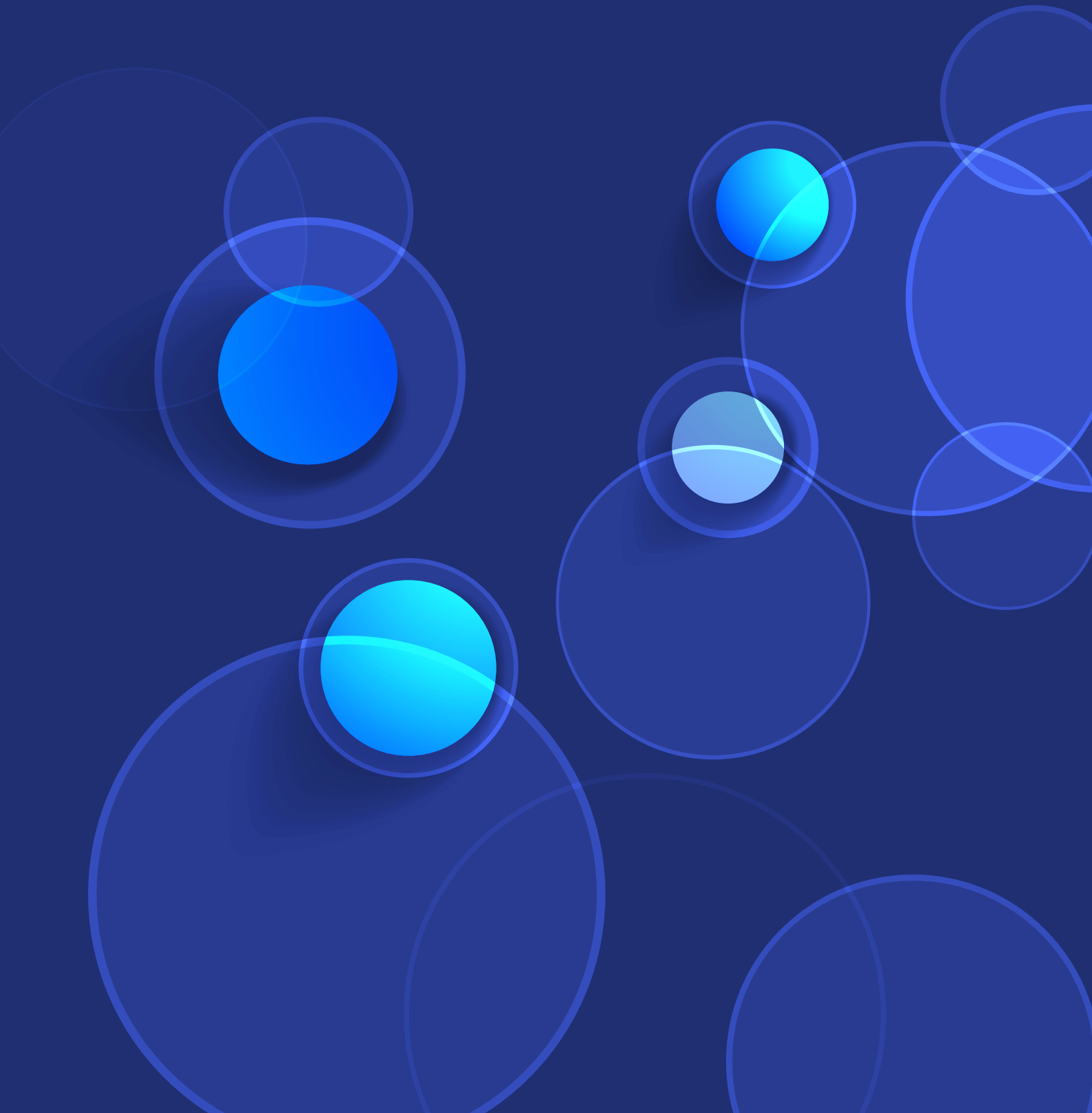
Phase	Action	
3b. Implementation plan/supply ■ ■ ■	Adherence to	☐ Agreed terms of supply and procurement
	Stakeholders	Initiated by: ☐ health technology developers Involved: ☐ hospital pharmacy ☐ payer organization ☐ Ministry of Health ☐ national procurement agency
	Sources of evidence	Not applicable
	Transparency	☐ Confidential
	Templates/tools	☐ No templates available
3c. Monitoring ■ ■ ■	Criteria	☐ Based on agreed terms
	Stakeholders	Initiated by: ☐ HTA agency and payer organization/Ministry of Health
	Sources of evidence	☐ Internal sources
	Transparency	☐ Mostly internal documentation
	Templates/tools	☐ No templates available
4. Managed exit (final year)		
Aim: to transparently execute the agreed upon terms (during the contracting phase) of exiting the MEA		
4a. Reassessment/re-appraisal (optional) ■ ■	Criteria	☐ New regulatory information (EPAR) from EMA or MHRA ☐ Availability of post-launch evidence from: a. significant published literature from clinical trials, observational data, consumption data or other data sources b. outcome-based MEAs ☐ New national clinical guideline ☐ Reaching outcome-based MEA targets before the anticipated time frame, e.g. reaching OS in an ongoing clinical trial when the outcome-based MEA is based on uncertainty of PFS translating to OS
	Stakeholders	Initiated by: ☐ HTA agency Involved: ☐ clinicians ☐ health technology developers ☐ Ministry of Health/payer
	Sources of evidence	☐ Updated reimbursement submission from health technology developer, including additional clinical information and evidence from outcome-based MEA ☐ Published literature ☐ Unpublished clinical evidence from post-launch evidence ☐ Recommendations from national scientific association
	Transparency	☐ Publication of reassessment report
	Templates/tools	☐ No templates available

Phase	Action	
4b. Re-negotiations (optional) ■ ■	Criteria	☐ New evidence on real-world effectiveness ☐ Adjustment to appropriate patient population based on RWD ☐ Agreement on new contract terms
	Stakeholders	<i>Initiated by:</i> ☐ Ministry of health or payer organizations <i>Involved:</i> ☐ Health technology developer
	Sources of evidence	☐ HTA reassessment report ☐ Adjusted economic analysis based on RWD
	Transparency	☐ Confidential
	Templates/tools	☐ No templates available
4c. De-investment (optional) ■ ■	Criteria	☐ Monitoring of early signs (e.g. FDA/EMA or other agency's withdrawal of product) ☐ EMA or United Kingdom/MHRA decision to withdraw the product ☐ MEA contractual terms not met ☐ Post-launch evidence (including patient-reported outcomes) shows insufficient real-world effectiveness
	Stakeholders	<i>Initiated by:</i> ☐ Payer organization or ☐ Ministry of Health <i>Involved:</i> ☐ health technology developer ☐ HTA agency ☐ clinicians/national medical scientific association ☐ patients ☐ patient organization ☐ communication with the public
	Sources of evidence	☐ New EPAR ☐ Post-launch evidence
	Transparency	☐ Announcement of de-investment with explanation in layman's terms
	Templates/tools	☐ No templates available

AIM: Association of Mutual Benefit Societies; ASCERTAIN: Affordability and Sustainability improvements through new pricing, Cost- Effectiveness and Reimbursement models to Appraise INnovative health technologies; EMA: European Medicines Agency; EPAR: European Public Assessment Report; EU HTAR: European Union Health Technology Assessment Regulation; EU-JCA: European Union Joint Clinical Assessment; FDA: United States Food and Drug Administration; HI-PRIX: Health Innovation Next Generation Payment & Pricing Models (project); HTACG: Coordination Group on Health Technology Assessment; IHSI: International Horizon Scanning Initiative; IT: information technology; MHRA: Medicines and Healthcare Products Regulatory Agency (United Kingdom); OB-MEA: outcome-based MEA; OS: overall survival; PFS: progression-free survival; PICO: population, intervention, comparator, outcome; RWD: real-world data; ZIN: *Zorginstituut Nederland* [National Health Care Institute].

Note: the colour scheme is the same as for the MEA complexity scale and indicates which steps are needed for each MEA type.

Future topics



Looking ahead, several areas warrant further exploration to ensure that the MEA principles evolve in step with broader health system developments. These include both structural and technological considerations to support evidence-informed, sustainable access models. Future topics for consideration include the following.

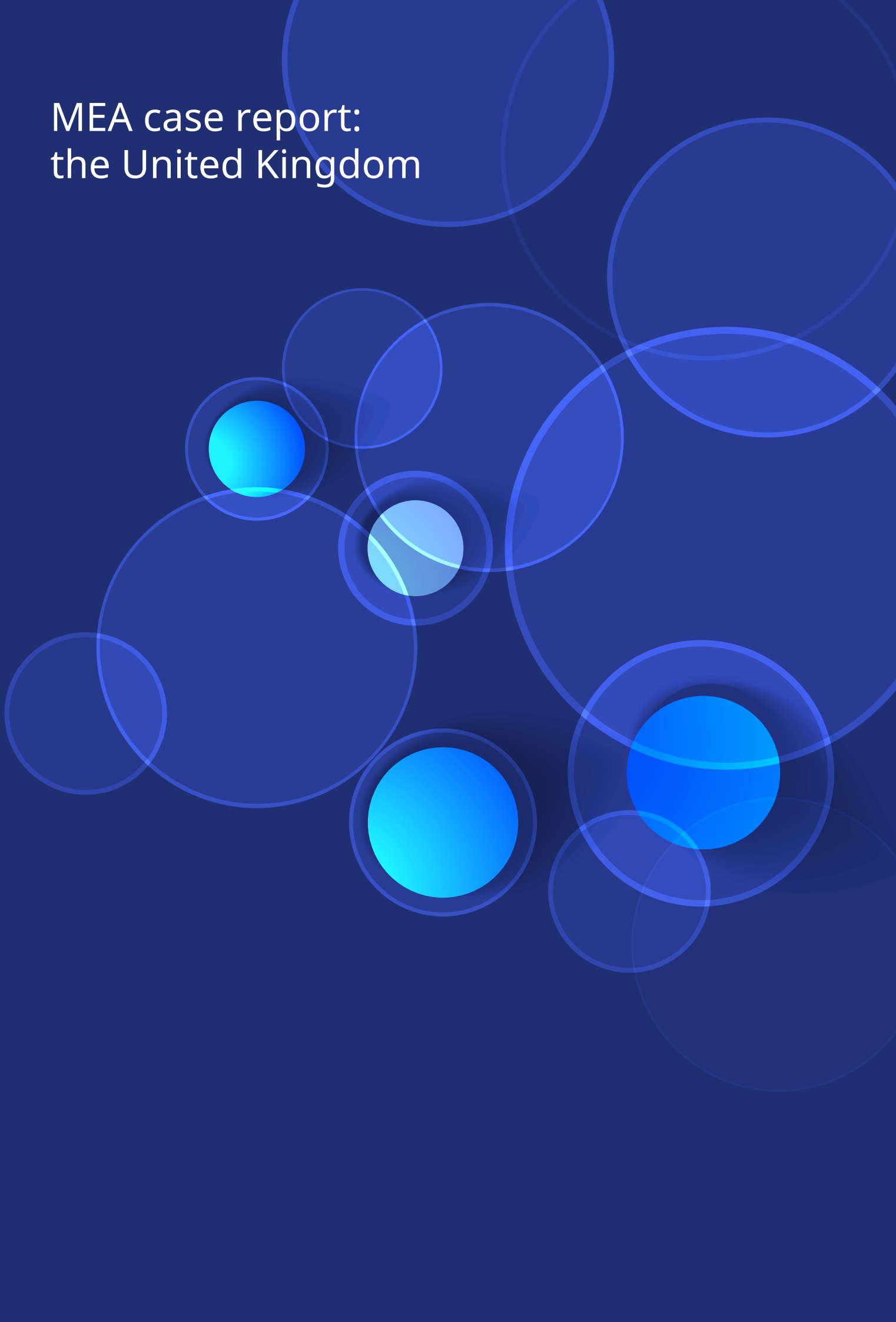
- **HTA/payer collaboration with extension to registry holders:** strengthening stakeholder ties between HTA bodies, payers and registry holders is essential to improve the utility of real-world data in assessing outcomes.
- **Transparency:** including the public availability and synthesis of MEA outcomes.
- **De-investment:** a clearly defined and structured process for de-investment is necessary to ensure consistency, patient continuity and appropriate communication when MEAs conclude or therapies are withdrawn from reimbursement.
- **MEA and appropriate use measures:** there is a growing need to connect MEAs with appropriate use measures to ensure that reimbursement is conditional depending on

outcomes and evidence of correct use in clinical practice.

- **Disruptive medicines:** the potential impact of disruptive medicines (e.g. one-time curative therapies or medicines with uncertain long-term outcomes) must be anticipated because these could challenge current MEA frameworks.
- **Learning health system:** as health care evolves towards a learning system, MEAs should accommodate novel innovations (including genomic sequencing, advanced screening and companion diagnostics) to support reimbursement aligned with evolving clinical insights.
- **Use of Artificial Intelligence (AI):** the integration of AI tools could enhance MEA administration, support earlier signal detection and optimize patient selection and follow-up processes.

These future topics for consideration present opportunities to develop MEAs and ensure that they remain fit for purpose in a rapidly evolving health-care landscape.

MEA case report: the United Kingdom



Organization of MEAs in the United Kingdom

National Health Service (NHS) provides and funds national health care for all citizens of the England and Wales. The aims of its operating framework include, but are not limited to, improving public health, health care quality, safety, outcomes and accessibility; reinforcing equity and equality; and providing value for taxpayers. NICE performs HTAs to provide guidance on medicines, products and treatments that are routinely funded by the NHS. NICE works with the English national payer, NHS England, to provide a range of MEAs that fall into two main categories: financial-based MEAs and coverage with evidence development (i.e. managed access agreements (MAAs)) (52). In practice, a mixed agreement is used that includes elements of financial-based MEAs and outcome-based MEAs.

NHS commercial framework: financial-based MEAs

NICE and NHS England have published a framework of potential commercial options that:

- supports companies in presenting a value proposition to NICE that is considered clinically effective and cost-effective within the relevant threshold;
- offers companies the opportunity to enter into complex, confidential agreements beyond a simple discount under specific circumstances and where an enhanced value proposition is presented to NICE that is at or below the lower end of the relevant clinical and standard cost-effectiveness threshold range;
- provides companies with a confidential commercial mechanism when NICE believes there is significant uncertainty surrounding the clinical effectiveness and, consequently, cost-effectiveness of a treatment and when there is plausible potential for that treatment to be clinically effective and cost-effective; and
- supports companies and NHS England to work together to identify confidential commercial solutions for NHS affordability challenges that may arise from otherwise clinically effective and cost-effective treatments.

A single commercial solution may be necessary to address more than one of the points outlined. The commercial options that can be considered fall into four broad categories:

- patient access schemes (PASs): simple and complex;
- commercial access agreements;
- MAAs; and
- budget impact schemes.

Coverage with evidence development: MAA

An MAA provides temporary patient access, funded by NHS England, for treatments that require further evidence of efficacy before a recommendation can be made for routine use. MAAs are initiated when NICE identifies and defines area(s) of uncertainty (i.e. in clinical or cost-effectiveness data of a novel drug following a technology appraisal) and assesses the feasibility of new data collection during granted patient access. The MAA outlines the process for collecting data to address and resolve uncertainties, thereby determining the long-term value of novel treatments. MAAs are crucial tools for balancing innovation with NHS budget constraints. They comprise two main parts: data collection agreements and commercial access agreements.

A drug is deemed eligible for an MAA if the uncertainties are well established, evidence collection is feasible and the data collection process will not impose an additional burden on the health system or patients. Further characteristics include ongoing trials or established data collection, a straightforward legal basis for data collection, a time frame of less than 2 or 5 years needed to overcome uncertainties and the willingness of stakeholders to undertake data collection.

MAAs are funded by the Cancer Drug Fund (CDF) (53) and the Innovative Medicines Fund (IMF) (54). The CDF and IMF aim to ensure access to novel treatments for which the medical need is high, thereby balancing the needs for cost-effectiveness and evidence-based decision-making. Each fund has an annual budget of £340 million, amounting to a total yearly budget of £680 million dedicated for novel medicines undergoing MAAs. Both the CDF and IMF use an expenditure control mechanism that prevents overspending.

- The CDF was integrated into the NICE process and relaunched in 2016 (55). It funds temporary patient access to oncological drugs during MAAs. All novel cancer drugs and novel licensed indications for cancer drugs are referred to NICE for technological appraisal.

- The IMF, launched in 2022, funds temporary patient access to non-cancer drugs during MAAs (56). It plays a crucial role in integrating advanced therapy medicinal products into the health system.

Principles underpinning the two managed access funds

- The CDF and IMF provide an equal opportunity for cancer and non-cancer patients.
- They target the most promising medicines that have significant resolvable uncertainty.
- Treatments must be plausibly cost-effective and priced responsibly during managed access.
- Treatments should stay in the funds for the shortest time necessary to resolve uncertainties – a maximum of 5 years.
- The entire eligible population can access treatment during the managed access period.
- HTA guidance is updated by NICE at the end of the managed access period.
- All patients can continue treatment in the event of a negative re-evaluation.
- The funds should never have to close to new entrants.

MEA phase 1: selection

Horizon scanning identifies technologies that are potential candidates for MAAs; it is undertaken by various stakeholders in England (United Kingdom), for example, NHS England, NICE and the Accelerated Access Collaborative. NICE uses the various outputs of horizon scanning to provide an overview of novel innovations, drugs and technologies that hold promise to improve patient care (57).

To become eligible for an MAA, the developer submits an evidence dossier to NICE, including a managed access proposal and a PAS (57). NICE requires an MAA proposal and feasibility assessment (which evaluates whether the technology is suitability for an MAA) to make a final recommendation on the routine use of the technology in the NHS (58).

NICE offers various evaluation programmes, but only the Technology Appraisal programme and Highly Specialised Technology programme are relevant to MAAs. This programme evaluates novel technologies based on clinical evidence of their safety and efficacy, as well as their plausible cost-effectiveness, resulting in a final draft guidance. There are three forms of technology appraisal:

- single (single technology for a single indication);
- streamlined approaches such as cost-comparison or cost-utility (single technology for a single indication, but with a shorter process that aims for accelerated patient access); and
- multiple technology appraisal (one or more technologies and for more than one indication).

Following an appraisal, NICE can make one of four recommendations in a final appraisal document (59). These are (i) recommended for routine use, meaning that NHS England will fund the drug; (ii) can be used during either managed access for technology appraisals and highly specialized technologies or an evidence generation period for medical technologies, diagnostics, early value assessments and interventional procedures; (iii) more research is needed; or (iv) not recommended for routine use. If the NICE appraisal committee recommends a health technology for routine NHS use, then it must be available to the population within the following 3 months.

Drugs are identified as potential CDF or IMF candidates at three different points during the appraisal process:

- when the company sends an evidence submission, including a managed access proposal, to NICE for technology appraisal;
- during the assessment phase (NICE may identify the drug as a potential CDF candidate); or
- at the first appraisal committee meeting (when the drug may be identified as a potential CDF candidate).

A crucial point in the process is that the NICE Technology Appraisal Committee must first rule out routine funding based on a simple financial agreement (confidential discount) before managed access in the CDF and IMF can be considered (see Figure 5).

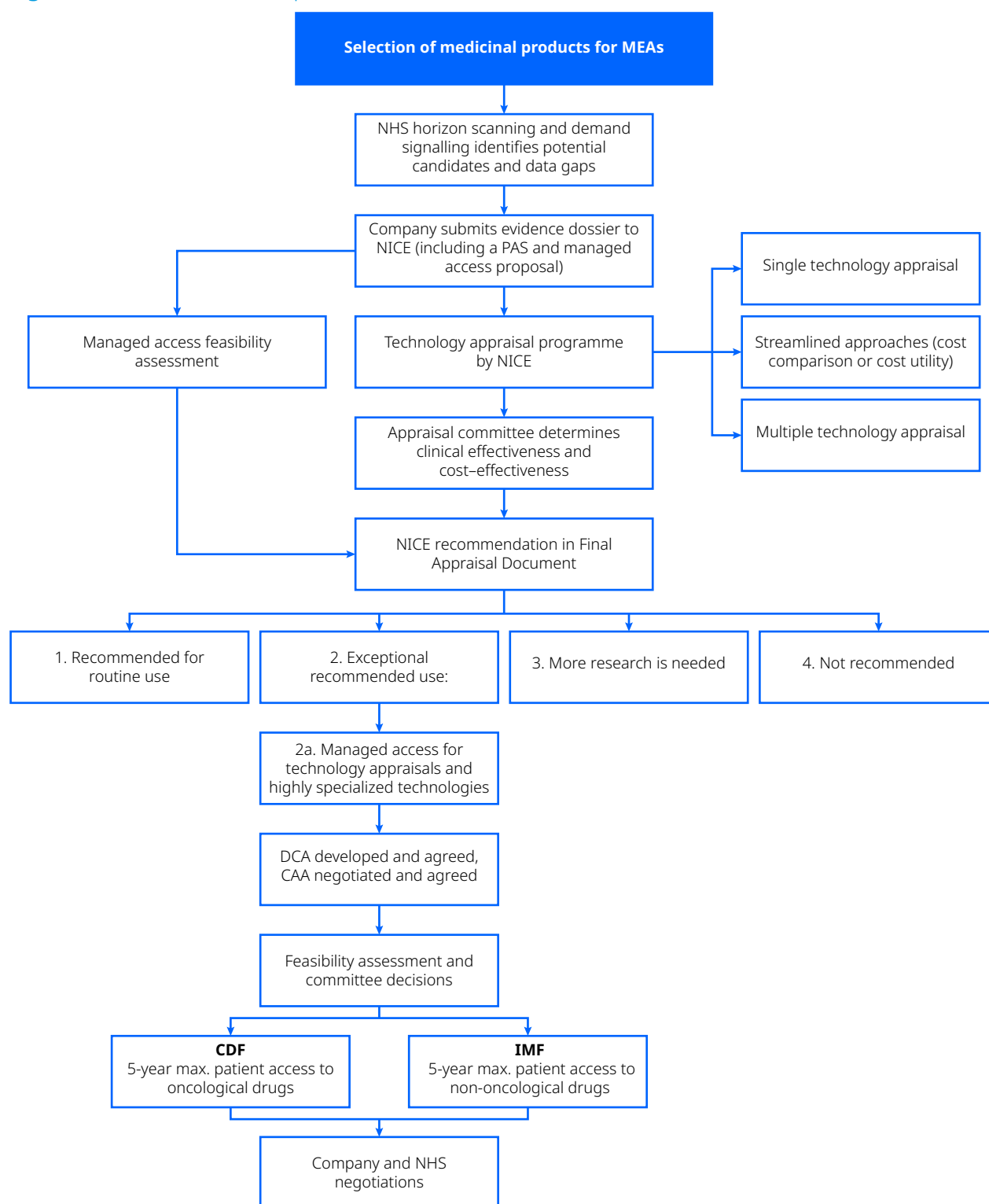
MEA phase 2: contracting

Various types of contracts underpin both financial-based MEAs and MAAs. A PAS forms part of a commercial access agreement and an MAA.

PAS

A PAS is the preferred option for companies that are developing a value proposition for evaluation by NICE. They provide a mechanism for companies to improve the cost-effectiveness of a treatment under evaluation beyond that driven by its list price. Unless a treatment is to be considered by NICE

at list price, companies are expected to include a PAS in their initial evidence submission to NICE to ensure that sufficient time is available for full

Fig. 5. Selection of medicinal products for MEAs

CAA: commercial access agreement; DCA: data collection agreement; max.: maximum.

consideration in advance of the NICE evaluation committee meeting.

The NICE Commercial Liaison Team advises NHS England on the feasibility of a PAS as set out in its *Procedure for the review of patient access scheme proposals* (60). This advice informs the final decision by NHS England on whether the proposed PAS can be considered as part of a NICE technology evaluation. NICE determines whether the level of discount being offered by the company corresponds to a clinically effective and cost-effective use of NHS resources.

Companies are expected to apply the following operational guidance to a PAS.

- The full costs to the NHS of any such arrangements should be included in the costs considered by the NICE evaluation committee.
- The PAS should be clinically robust, clinically plausible, appropriate and monitorable (e.g. if it is a responder scheme, then there must be a relatively straightforward way to measure a patient's clinical response).
- Any PAS should be operationally manageable for the NHS without unduly complex monitoring or disproportionate additional costs and bureaucracy. Any burden for the NHS should be proportionate to the benefits of the PAS for both the NHS and patients. It is reasonable for NHS England to take operational management into account when considering the viability of individual PAS proposals, and it is likely to prioritize PAS proposals that deliver the most significant benefits to patients (e.g. by enabling the NHS to address a previously unmet need).
- Clarity is required on the exact duration of any agreement and the circumstances in which it might be terminated.
- A PAS should be consistent with existing financial flows within the NHS and with commissioning arrangements (e.g. payers must be able to calculate the effective price for their patient population so the costs and savings accrue to those services making commissioning and treatment decisions).
- The NHS in England and Wales must be consulted on PAS proposals, particularly where these involve data collection beyond that associated with the conventional purchase of medicines (e.g. patient numbers or monitoring/recording a patient's condition over and above that required for standard management). The Commercial Liaison Team's process includes arrangements for consultation with the NHS.

There are two types of PAS.

- A simple PAS (confidential) provides a fixed price or percentage discount on the list price, which is applied at the source. It is always the preferred option because it requires less monitoring by all parties and minimize the administrative burden on NHS organizations. It also ensures that where value added tax is incurred, it is applied at the lowest level for the effective net price. Review and approval are more rapid for a simple PAS than for a complex PAS.
- A complex PAS (transparent) is not confidential. By definition, it will involve a more complex reimbursement proposal that, in turn, will be more complex to administer within the NHS. The requirement for transparency is to ensure that the administrative burden and cost to the NHS of implementing such PASs are minimized, which in turn helps to ensure that the value of the treatment, as determined by NICE, is achieved. If a company chooses to propose a complex PAS, then it must justify this with a strong rationale for the PAS and indicate how the associated risks will be shared equitably between the NHS and the company. The proposal must also account for the consequences for any related value added tax to ensure that the NHS can realize the full financial benefits of the PAS. NICE Commercial Liaison Team's advice on the feasibility of implementing a proposed complex PAS is inevitably determined via a more-involved process, including consultation with the NHS and a feasibility review of commercial arrangements, to ensure that the full benefit can be realized.

Commercial access agreements

Commercial access agreements are confidential and are agreed by NHS England. The default arrangement of offering a PAS (simple or complex) is always available to companies. NHS England encourages companies to engage at an early stage in commercial options, particularly when they believe that a commercial access agreement is needed, to mitigate delays later in the evaluation process. Early discussions through commercial surgeries facilitate discussions about the commercial case, including any data requirements, and provide an initial indication of the potential flexibilities available. Table 2 shows example formats for a commercial access agreement (or complex PAS), but each commercial access agreement is considered on a case-by-case basis.

Table 2. Example scheme types for a commercial access agreement arrangements or complex PAS^a

Scheme type	Description
Budget cap	Maximum budget impact for a product (or products) beyond which a central rebate is payable
Price/volume agreement	A price is agreed for a set volume of patients and then either reductions are staged based on additional patient numbers or the company pays back the full amount (similar to a budget cap)
Cost sharing	The company funds the initial cost of therapies, such as by offering the first month for free
Stop/start criteria	Rules on eligibility criteria that determine when patients should start and stop therapy
Outcomes-based agreement and payment by results	A discount or rebate is applied if a product does not perform as expected or for non-responders
Indication-specific pricing	Differential prices for a given medicine based on the differences in clinical effectiveness and cost-effectiveness across indications

Note: this list is not exhaustive, and a combination of scheme types can be applied.

^a A commercial access agreement and complex PAS differ in relation to the transparency of the agreement and to distinction between central and local implementation, transaction, and monitoring.

Specific circumstances may render the launch of a product particularly challenging or commercially unviable when based solely on a PAS, and NHS England will consider a commercial access agreement in such circumstances. An example is when the company proposes an enhanced value proposition that would normally be at or below the lower end of the standard NICE cost-effectiveness threshold range. Greater flexibility is available for value propositions at even greater levels of cost-effectiveness, plus any applicable quality-adjusted life-year weighting.

Where an indication-specific arrangement is agreed for a medicine, the indication-specific price under this arrangement will then be the price used in any subsequent evaluations for the same indication when the medicine is used as a comparator. There may be specific circumstances beyond those considered here that justify bespoke commercial flexibilities, and these are assessed on a case-by-case basis.

MAAs

MAAs have two key components (61): a data collection agreement to mitigate clinical uncertainty (as defined by the NICE evaluation committee); and either a commercial access agreement or a PAS (simple or complex).

MAAs represent an interim commissioning position, with a future date committed to for re-evaluation, which may result in routine commissioning. They are, therefore, time-limited agreements.

Companies are required to agree to offer the treatment at a cost-effective price for the duration of the MAA. Each MAA includes exit clauses, including the obligation to maintain funding and existing patient access should any reassessment result in a negative decision. It should be noted (as with the CDF and IMF) that when exiting the MAA the price of the product may be higher or lower, depending on the re-evaluation outcome. A key constraint to overcome when considering the possibility of a MAA is whether data can be collected on the relevant health outcomes and the collected data can resolve the uncertainties.

Data collection agreements

This type of agreement is developed by NICE in collaboration with relevant stakeholders, including clinicians, patient groups, academics and data custodians. They specify the clinical data required to resolve uncertainties and define applicable data sources, such as real-world evidence or ongoing clinical trials, and outline the framework for data analysis. The agreement further discloses stakeholder responsibilities and the relevant patient population, type of clinical assessment and data to be collected and analysed.

Data are typically collected for around 2 years, but the period is determined on a case-by-case basis by NICE. CDF and IMF have a maximum data collection time frame of 5 years, with review points set to track progress. Pharmaceutical companies fund the data collection, management and analysis, whereas NICE oversees operational aspects. NICE

collaborates with the pharmaceutical company to ensure the feasibility and minimize the burden of data collection. Both CDF and the IMF rely on the agreement to ensure the timely collection of evidence during patient access. Failure to collect data in accordance with the agreement may result in negative final guidance after re-evaluation or in termination of the re-evaluation.

MEA phase 3: implementation

Data collection during managed access must be proportionate to the uncertainties identified by the NICE appraisal committee. Preferred sources of data include phase III clinical trial data, the NICE real-world evidence framework, and trial or registry data (i.e. Public Health England). The Managed Access Investment group reviews and publishes quarterly reports on the number of funded patients. The NICE committee aligns reviews with appraisal timelines and may withdraw funding if significant delays occur during the data collection process.

Several NHS England systems further support data collection, including High Cost and Cancer Drug Prior Approval System (Blueteq), electronic chemotherapy prescribing, the Systemic Anti-Cancer Therapy dataset and the CDF notification system. These systems can be used in combination to complement data collection methods.

Blueteq provides a digital platform that facilitates the approval, tracking and management of high-cost drugs by offering a complete patient history, centralizing documentation and ensuring real-time access to vital data for clinicians and stakeholders.

All chemotherapy providers in the NHS use electronic prescribing (e-prescribing). E-prescribing ensures safe, efficient and standardized cancer treatment. It also reduces prescription errors, minimizes drug waste and reinforces the use of evidence-informed chemotherapy regimens. To ensure patient access, e-prescribing must be performed for all drug prescriptions, cover all administration routes and submit complete data for each CDF-funded drug.

The Systemic Anticancer Therapy dataset collects data on all patients receiving systemic anticancer therapy and provides real-time information on trends in cancer treatment. The dataset facilitates financial planning, policy development and specialized cancer commissioning by the NHS. It is the preferred method for routine chemotherapy data collection owing to its established data structure, data protection and efficient monitoring abilities. The CDF notification system maintains a centralized list of approved drugs and criteria, which streamlines access to CDF-funded treatments. The NHS regularly updates the CDF drug list and ensures compliance (non-adherence may affect reimbursement).

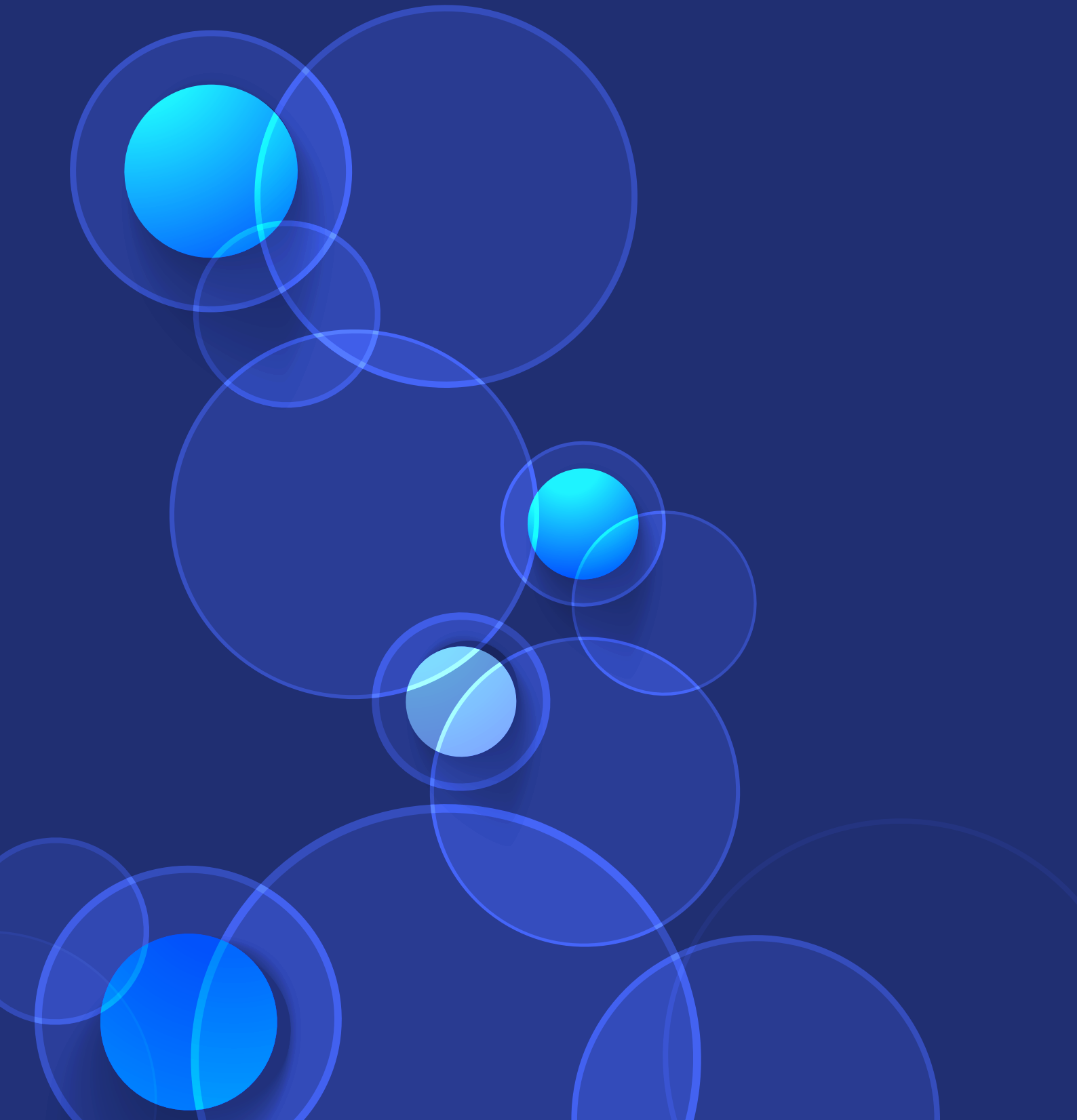
MEA phase 4: managed exit

After the data collection period, NICE reappraises the topic if its appraisal committee issues a positive recommendation in the final draft guidance. The drug then transitions to routine NHS commissioning, with interim funding (standard for CDF, optional for IMF), before full integration into NHS England.

If the recommendation is negative, then routine funding ceases, with no future access. Existing patients must continue the treatment at the pharmaceutical company's expense until it is deemed clinically unnecessary by both clinicians and the patient. The requirement to continue funding in the event of a negative recommendation is part of the MAA and commercial access agreement that companies must agree to before they can enter managed access.

Data collection agreements must have previously defined exit strategies, including potential price adjustments, and alignment with NHS policies and procedures. NHS England and NICE conduct interim reviews to assess the progress of data collection. If insufficient data are gathered to address uncertainties and this cannot be rectified, then the agreement may be terminated before re-appraisal can take place.

MEA case report: Netherlands (Kingdom of the)



In Netherlands (Kingdom of the), novel drugs are classified as outpatient or inpatient pharmaceuticals. The health system is State governed and financed through private insurance, premiums and taxpayer contributions (62). Under the Health Insurance Act, all residents are entitled to a basic health insurance package (63). To support cost-effective and equitable health care, *Zorginstituut Nederland* [National Health Care Institute] (ZIN) conducts HTAs to evaluate novel treatments before reimbursement. After market entry, ZIN implements “appropriate package management” to ensure that only drugs offering added value to society are reimbursed, with an emphasis on transparent decision-making (64). One key package criterion is the “state of science and practice” (63), which reflects whether a drug provides proven benefit and aligns with current clinical standards.

Clinical uncertainty despite a high medical need for a novel drug may require conditional reimbursement, under which a treatment becomes accessible to the population under controlled and

temporary conditions. In Netherlands (Kingdom of the), there are two types of managed entry pathway:

- *Voorwaardelijk toegelaten zorg* [conditionally approved care] (VT), as operated by ZIN and executed by the Ministry of Health, Welfare and Sport (65); and
- Drug Access Protocol (specifically for cancer medicines) and Orphan Drug Access Protocol, as coordinated by *Zorgverzekeraars Nederland* [Health Insurers of the Netherlands] (66,67).

Two upcoming articles describe in detail the processes of each programme: *Coverage with evidence development for medicines with insufficient evidence of clinical benefit: experience from the Netherlands* describes the VT operated by ZIN (68) and *Stakeholders’ experiences with a clinician-led access program linking evidence generation and reimbursement for precision cancer treatments: the Drug Access Protocol in the Netherlands* describes the Drug Access Protocol (69).

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² All references were accessed on 7 November 2025.

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Annex 1. Organizations that participated in the deliberation process

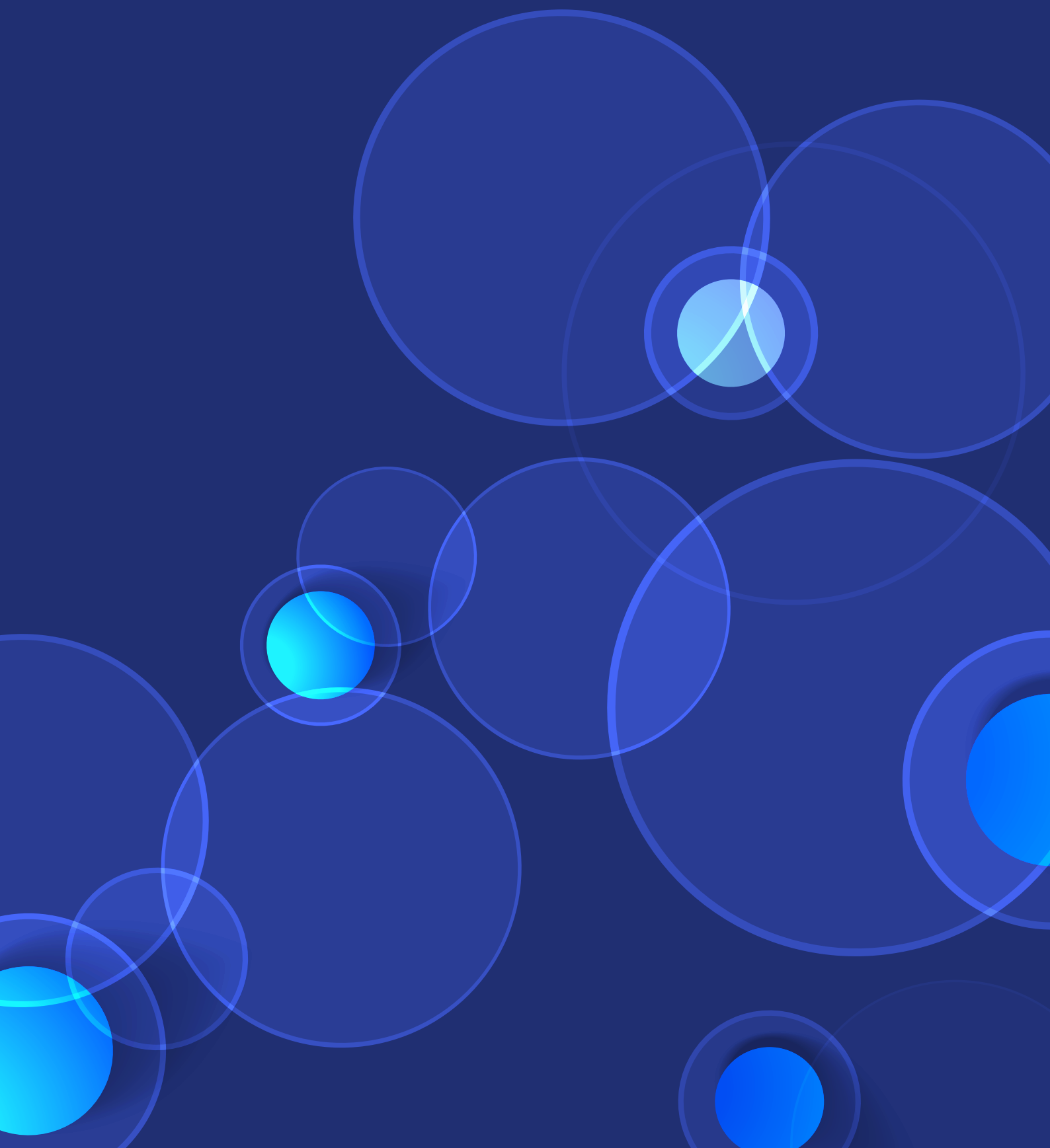


Table A1.1 shows a list of organizations that participated in the deliberation process.

Table A1.1. Organizations that participated in the deliberation process

Country	Institution	Type of feedback
Austria	Austrian Public Health Institute/WHO Collaborating Centre for Pharmaceutical Pricing and Reimbursement/Secretariat of the Collaborating Centre	Informal conversation
Belgium	National Institute for Health and Disability Insurance	Informal conversation
Denmark	Amgros	Informal conversation
Germany	Federal Ministry for Health	Informal conversation
Ireland	Health Service Executive	Informal conversation
Italy	Italian Medicines Agency	Informal conversation
Netherlands (Kingdom of the)	<i>Zorginstituut Nederland</i>	Informal conversation
Norway	Nordic Pharmaceutical Forum	Informal conversation
Sweden	Dental and Pharmaceutical Benefits Agency	Informal conversation
European Union	Organisation for Economic Co-Operation and Development	Informal conversation
European Union	Affordable Medicines Europe	Written feedback
European Union	RWE4Decisions	Multistakeholder round-table discussion
European Union	European Confederation of Pharmaceutical Entrepreneurs	Informal conversation

Annex 2. Overview of all feedback

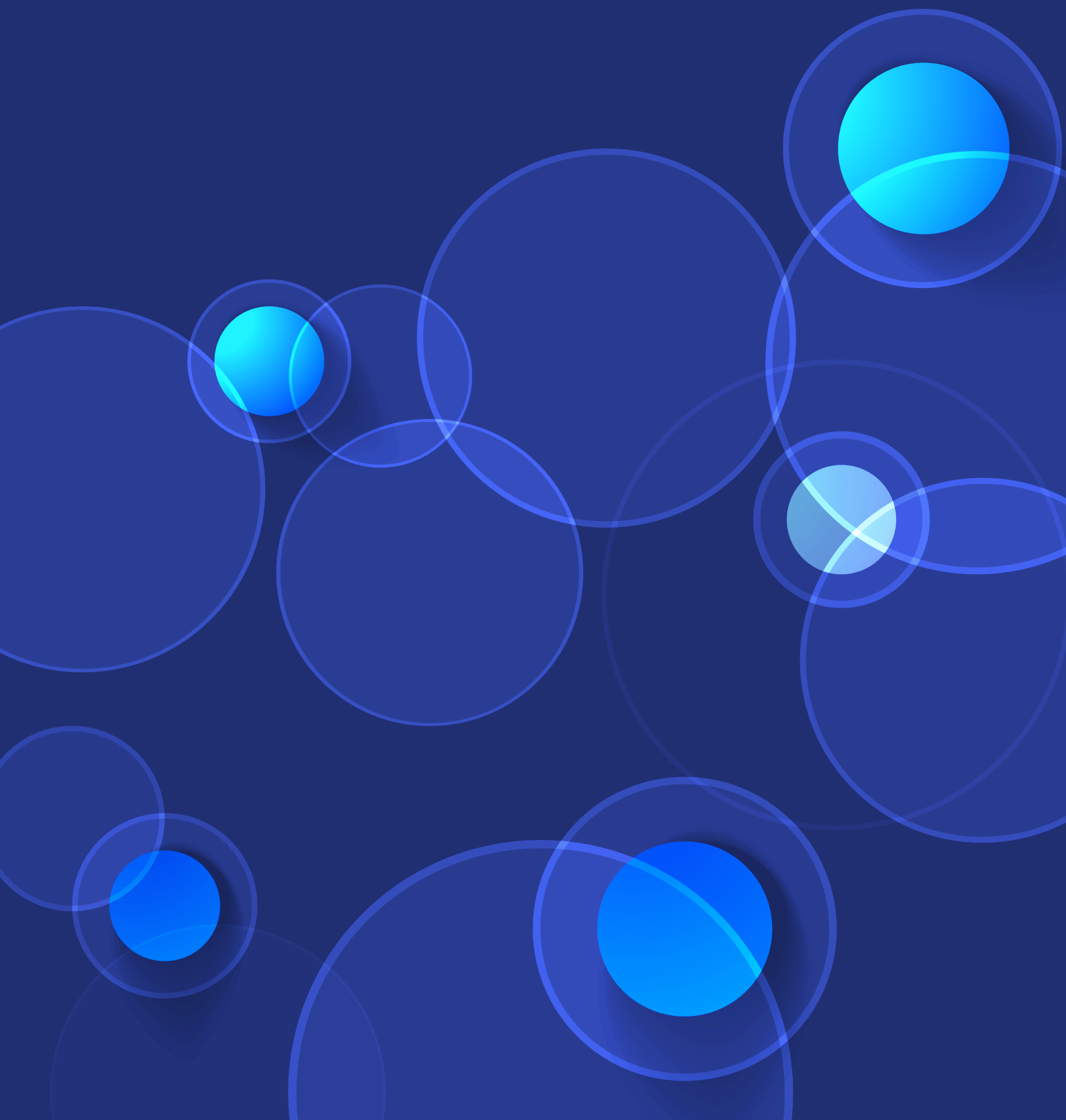


Table A2.1 shows the feedback received on draft 1 in round 1 and Table A2.2 shows the feedback received on draft 2 in round 2.

Table A2.1. Feedback received in round 1

Type of feedback	Comments
Contracting	• Implement standard templates for budget capping/pricing structures, centre financial incentives in MEAs • Consider the consequence of price-volume agreements/budget thresholds for supply incentives and access • Manage economic and clinical expectations to benefit clinical analysis and robust comparisons at the start
Data	• Encourage data-sharing for international coordination in collecting real-world evidence (beneficial for rare indications) • Discourage use of the AIM fair price calculator given its cost-plus nature (does not in-centivize clinical development in unmet need areas nor does it inform MEA principles; contradicts value-based pricing) • To obtain reimbursement, stakeholders must fill in all required data for the registry
Implementation	• For complex MEAs, consider the trade-off between long-term benefit and ease of implementation
Managed exit	• Clarify the reassessment criteria for pricing adjustments and possible de-investment mechanisms • Define structured terms and enforceable exit indicators • Clarify consequences for patients already on the treatment procedure in case of a negative reimbursement decision
Pricing and procurement	• Centralize towards national procurement for high-cost medicines • Differentiate MEAs with simple discounts • MEAs are used as a caching system to resolve uncertainties • Decrease the financial focus in MEAs, and focus more on the clinical parameters
Selection	• Improve budget forecasting and societal value estimation at the selection stage • Consider availability of the principal sources needed for the selection phase (EPAR, real-world data, clinical studies) • Clarifying the objectives of MEAs to discourage the promotion of “simple” MEAs because objectives differ for each drug
Stakeholder relations	• Increase cross-learning on agreement details to increase trust between stakeholders • Bridge any disconnect in incentives between stakeholders (HTA cost-effectiveness, payers affordability) • Ensure that all stakeholders are represented in MEA phases (patients, data experts, industry, health technology developers) • Develop a structure to track MEA progress, e.g. regular evaluations such as stakeholder meetings
Transparency	• Implement increased transparency on MEA design, implementation and stakeholder roles; maintain confidentiality on pricing • Consider data-sharing challenges at patient level • Enable partial transparency for the purposes of competition; inform affected stakeholders of pricing
Overall feedback	• Make MEA principles a living document • Replace the traffic light system with alternative visuals to avoid misinterpretation as a go/no-go system

MEA: Managed Entry Agreement; AIM: Association of Mutual Benefit Societies; EPAR: European Public Assessment Report.

Table A2.2. Feedback received in round 2

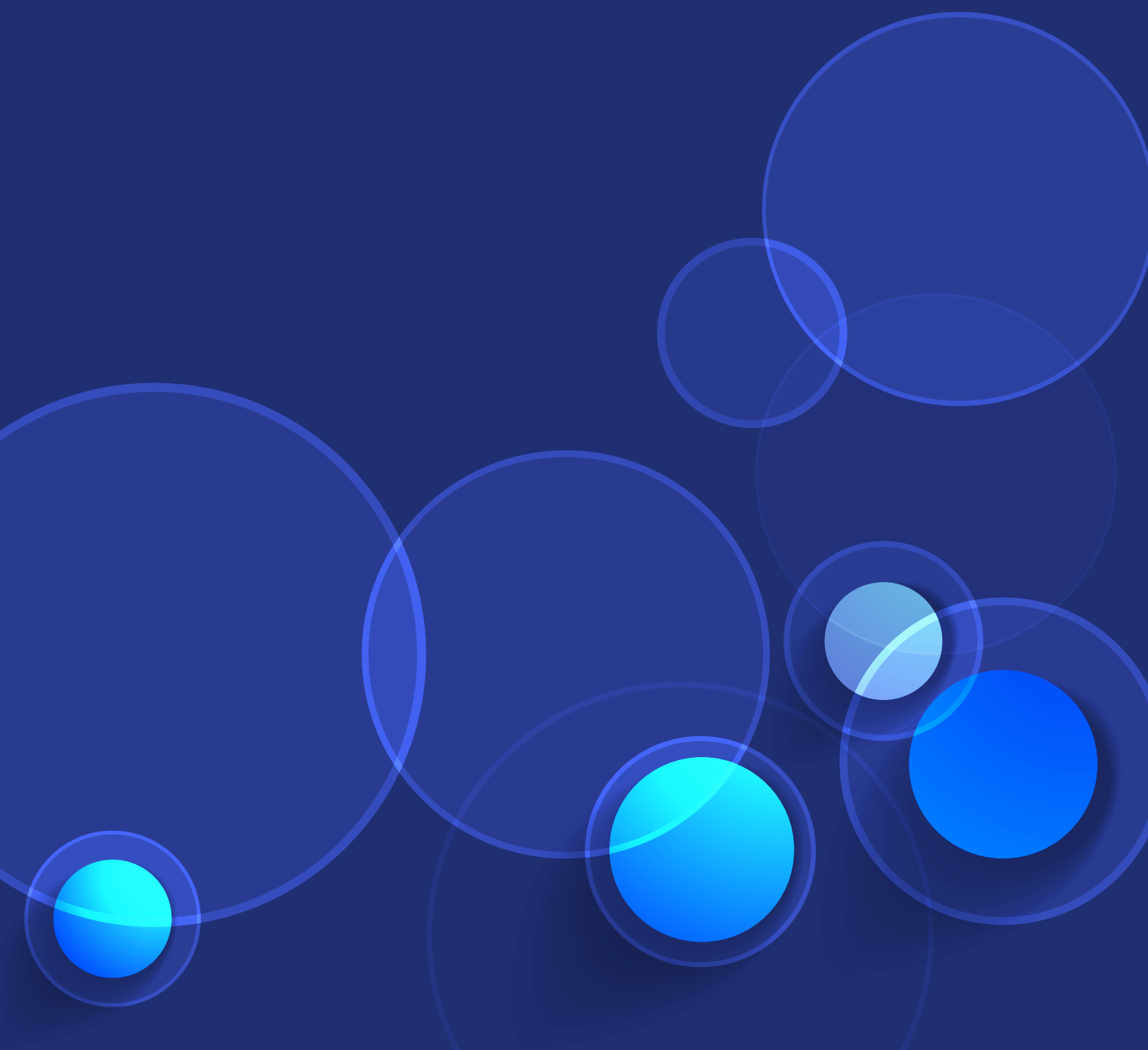
Type of feedback	Comments
Contracting	• Explicitly state the aim of the specific MEA, e.g. the clinical uncertainty being addressed • A 2- to 5-year time frame may be too restrictive • Add the principle of enforcing a regular evaluation of the entire process, including how many companies fulfill the condition, amount of payback yearly and the amount of resources from the health systems to MEAs
Data	• Set up end-points, with a preference for simple patient-relevant end-points (e.g. overall survival) over observer-dependent, non-validated surrogate end-points (e.g. response rate) • Ensure that the outcomes of data collection targets are pivotal to the final decision-making (e.g. uncertainty about cost-effectiveness) • Define the expectations for outcome measurements, including patient-reported outcomes and quality-of-life measures
Managed exit	• Clarify the exit terms to determine reimbursement or patient access in case of a negative recommendation • Allow for the possibility of contract renewal after the MEA ends
Pricing and procurement	• Missing financial elements include net prices paid, negotiated discounts, research and development costs, and savings achieved • Include rebates and price variations (e.g. reverse, per patient)
Selection	• This stage should occur before an HTA of uncertainty because without registries/data sources, the lead times and clinician consultation are first required to plan for what evidence is needed • MEAs should be used only when they offer net societal benefit by accounting for access, uncertainty, expenditure and health opportunity costs; retain the option to reject the product, if necessary • Include the potential for patients with a high unmet need and weigh people-centred outcomes against uncertainty level
Stakeholder relations	• Include entities acting on a manufacturer's behalf (e.g. local branches, wholesalers, joint ventures) • The payer and health technology developer should agree on data collection terms • Short MEAs may favour manufacturers by securing early access and making it difficult for payers to withdraw coverage later, raising concerns about negotiating leverage in the long term
Transparency	• Make the first MEA principle "open and transparent reimbursement as the default", in line with resolution WHA72.8 on improving the transparency of markets for medicines, vaccines, and other health products (1) • Stakeholders should publish a selected product list to support candidate identification • Incorporate financial transparency in future outputs to improve affordability and equitable access • Clarify whether data collection will be shared with the manufacturer
Other	• Simply removing the red colour from the flow chart may not be sufficient owing to the strong negative reaction to the original traffic light system; the recommendation is to use NMP tones/colours instead of traffic light colours

NMP: WHO Regional Office for Europe's Access to Novel Medicines Platform.

Reference

1. Seventy-second World Health Assembly, 28 May 2019. Agenda item 11.7: Improving the transparency of markets for medicines, vaccines, and other health products. Geneva: World Health Organization; 2019 (WHA72.8; <https://iris.who.int/handle/10665/329301>), accessed 7 November 2025.

Annex 3. MEA checklist 1: potential selection criteria



The following questions (Q) aim to encourage the use of the simplest MEA option as the first choice to ensure the minimal burden.

Q1. Is there a significant high unmet medical need in the therapy area that would justify deviation from the standard routes to evaluation and commissioning?

Q2. Is the eligible patient population tightly defined (via confirmed diagnosis) and not heterogeneous?

Q3. Are any alternative technologies being developed that are due for marketing authorization within 1 year of the product being launched?

Q4. Is the clinical outcome clearly defined, measurable and unambiguous, with the ability to convert/map into quality-adjusted life-years?

Q5. Do the clinical benefits of using the technology represent a step change in addressing clinical unmet need or improving patient-relevant outcomes such as quality of life, functional status or patient experience?

Q6. Can the uncertainty be plausibly resolved via data collection?

Q7. Are these technologies available in geographical locations where MEA is considered and where patients live?

Q8. Can the company reliably supply the product to all patients being considered for the outcome-based MEA

Q9. Is there a robust source of data (both financial and clinical) that is routinely collected or can be collected without placing an undue administrative burden on the health-care service?

Q10. Is the exit strategy for the MEA clearly defined and acceptable to clinicians, patients and commissioners?

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